Deep SURE for Unsupervised Remote Sensing Image Fusion

Han V. Nguyen, *Member, IEEE*, Magnus O. Ulfarsson, *Senior Member, IEEE*, Johannes R. Sveinsson, *Senior Member, IEEE*, and Mauro Dalla Mura, *Senior Member, IEEE*

***Abstract*— Image fusion is utilized in remote sensing (RS) due to the limitation of the imaging sensor and the high cost of simultaneously acquiring high spatial and spectral resolution images. Optical RS imaging systems usually provide images of high spatial resolution but low spectral resolution and vice versa. Therefore, fusing those images to obtain a fused image having both high spectral and spatial resolution is desirable in many applications. This article proposes a fusion framework using an unsupervised convolutional neural network (CNN) and Stein's unbiased risk estimate (SURE). We derive a new loss function for a CNN that incorporates a backprojection mean square error (MSE) with SURE to estimate the projected mse between the fused image and the ground truth. The main motivation is that training a CNN with this SURE loss function is unsupervised and avoids overfitting. Experimental results for two fusion examples, multispectral and hyperspectral (MS–HS) image fusion and multispectral and multispectral (MS–MS) image fusion, show that the proposed method yields high-quality fused images and outperforms the competitive methods. Codes are available at https://github.com/hvn2/Deep-SURE-Fusion.**

***Index Terms*— Image fusion, multispectral and hyperspectral (MS–HS) image fusion, pansharpening, remote sensing (RS), Sentinel 2 sharpening, Stein's unbiased risk estimate (SURE), unsupervised convolutional neural networks (CNNs).**

I. Introduction

OPTICAL remote sensing (RS) involves capturing reflection from the Earth's surface using imaging sensors mounted on airborne or spaceborne systems [1]. Those optical RS imaging systems usually provide spectral and spatial resolution varying from a few to several hundred meters. For example, a hyperspectral image (HSI) comprises several hundred narrow bands and is useful for applications such as classification [2] and object detection [3], because the rich spectral information allows discriminating material signatures. Another example is multispectral imaging systems that provide a multispectral image (MSI) and a panchromatic (PAN) image. Those images are useful for crop mapping [4] and environmental monitoring [5]. One common issue of an RS imaging sensor is a tradeoff between spectral resolution and spatial resolution. This means that an HSI has a higher spectral resolution than an MSI and a PAN image, but it has a lower spatial resolution, and it is noisier than both of them. Therefore, to maximize the utilization of both spectral and spatial information of HSI, MSI, and PAN image, a straightforward and economical way is to merge those images, resulting in a fused image having both high spectral and spatial resolutions. This process is known as image fusion [6], which can be considered a preprocessing technique for many subsequent applications [7], [8], [9].

There are several image fusion problems in optical RS depending on the high resolution (HR) and low resolution (LR) images to be fused. Pansharpening [10], [11] and hyperspectral pansharpening [12] involve fusion of a PAN image with an MSI and an HSI, respectively. Hypersharpening [13] is the fusion problem where the spatial source (i.e., HR image) can be composed of several bands. The fusion problems addressed in this article are all instances in the framework of hypersharpening.

Generally, optical RS fusion can be categorized into three approaches. The first approach consists of classical component substitution (CS) [14], [15], [16] and multiresolution analysis (MRA) [17], [18], [19] methods. The second approach contains the model-based methods, and the third approach includes deep learning (DL)-based methods. The CS and MRA methods have been extended for hypersharpening [20], multispectral and hyperspectral (MS–HS) fusion [21], and Sentinel 2 (S2) sharpening [22].

Model-based methods usually formulate the fusion problem as an ill-posed inverse problem. Solving this inverse problem requires prior knowledge (i.e., image prior) about the estimated images to constrain the searching space. Since RS images have a high spectral correlation, they have a low-rank characteristic, i.e., it is possible to represent them in a low-dimensional space. Several RS image fusion methods have exploited the low-rank characteristic, such as in MS–HS fusion [23], [24], [25], [26], [27], S2 sharpening [28], [29], [30], [31], and pansharpening [32], [33]. Other model-based methods include total variation (TV) methods [34], [35], nonlocal

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similarity methods [36], [37], and methods based on sparse representation [38], [39]. The main limitation of those model-based methods is high computation load and sensitivity to tuning parameters.

Most DL-based methods learn a complex network mapping function trained by using a big dataset containing observation and ground-truth pairs (supervised learning). Representative DL-based methods are [40], [41], and [42] for MS–HS fusion, [43], [44], and [45] for pansharpening, and [46], [47], and [48] for S2 sharpening. The supervised DL-based methods [40], [41], [42], [43], [44], [45], [46], [47], [48] have demonstrated their efficiency and obtained state-of-the-art results. However, the performance of supervised DL-based methods depends heavily on the training data. It is hard to acquire the observation and ground-truth image pairs in RS due to the high cost and sensor limitations. Thus, the training datasets are artificially synthesized using a sensor degradation assumption. Training a network with a synthesized dataset may not be correct for real applications [49]. Note that there are also some unsupervised DL-based RS fusion methods [50], [51], [52].

This article proposes an RS fusion framework using an unsupervised convolutional neural network (CNN) and Stein's unbiased risk estimate (SURE) [53], [54]. Inspired by the deep image prior (DIP) [55] theory stating that the structure of a CNN provides a good image prior for an image, we further investigate SURE to improve the DIP results. Specifically, we optimize (train) a CNN using a new SURE-based loss function, which is an unbiased estimate of the mean square error (MSE) of the projected difference between the fused image and the ground truth. The method is unsupervised since it does not require any ground truth, and we demonstrate that the fusion results are improved. The main contributions of this article are summarized as follows.

1) We propose a general RS image fusion framework using an unsupervised CNN and SURE. Interestingly, the method can be considered a hybrid DL and model-based method since it uses an image prior provided by the CNN structure to deal with the inverse problem and a DL optimizer to solve the optimization problem.
2) We propose a new loss function in which a linear inverse filter that maps an LR image to its HR space is incorporated with SURE. This SURE-based loss function avoids overfitting when training a CNN with a DIP-based loss function and improves the fusion results.
3) We address two common cases of RS image fusion using the proposed method, i.e., when the noise level is high (e.g., in MS–HS fusion) and when the noise level is low (e.g., in S2 sharpening). Experimental results show that the SURE loss function significantly improves the fusion results in the low noise case and efficiently avoids overfitting in the high noise case.

This article is organized as follows. Section II presents the problem formulation and related work, and the proposed method is described in Section III. We provide the experimental results MS–HS and multispectral and multispectral (MS–MS) fusion in Sections IV and V, respectively. Finally, the conclusions are drawn in Section VI.

A. Notation

We denote a matrix with a bold, uppercase letter, e.g., $\mathbf{X} = [\mathbf{x}_i]_{i=1}^n$, where its ith column is $\mathbf{x}_i$. A vector is denoted as a bold, lowercase letter, e.g., $\mathbf{x} = [x_i]_{i=1}^n$, where its ith element is x_i. An image is represented in either a matrix by vectorizing a band as a column or as a vector by stacking each column of the image matrix on top of each other. The vectorizing operator is denoted as $\text{vec}(\cdot)$, i.e., $\mathbf{x} = \text{vec}(\mathbf{X})$.

II. Problem Formulation and Related Work

A. Problem Formulation

In this article, we address the fusion problem of MSI/HSI acquired by the same optical RS modality. It means that those images are registered and are only different in terms of spatial and spectral resolution. The observations are an LR image $\mathbf{Y}$ of $M = h \times w$ pixels and C-bands, and a guided image $\mathbf{G}$ of $N = H \times W$ pixels and c-bands, where h, w, H, and W are spatial sizes of the LR and guided images. Here, the LR image $\mathbf{Y}$ has a lower spatial resolution, but more spectral bands than the guided image $\mathbf{G}$, i.e., we assume that $H = d \times h$, $W = d \times w$, and $c < C$, where d is a positive integer. Our problem is to estimate a fused image $\hat{\mathbf{X}} \in \mathbb{R}^{N \times C}$ from the following model:

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \boldsymbol{\epsilon} \tag{1}$$

$$\mathbf{G} = \mathbf{X}\mathbf{R} + \mathbf{N}_g \tag{2}$$

where $\mathbf{H}$ is a block diagonal matrix, i.e,

$$\mathbf{H} = \mathbf{M}\mathbf{B} = \text{bdiag}(\mathbf{H}_1, \mathbf{H}_2, \ldots, \mathbf{H}_C)$$

where each diagonal element can be written as $\mathbf{H}_i = \mathbf{M}_i\mathbf{B}_i$, $i = 1, 2, \ldots, C$, which is spatial degradation operating on the ith band and is composed of a spatial blurring filter, $\mathbf{B}_i$, and a downsampling matrix by d_i, $\mathbf{M}_i$. Under a circular boundary condition, $\mathbf{B}_i$ is a block-circulant–circulant-block matrix, and the blurring filter can be efficiently implemented using the fast Fourier transform (FFT). $\mathbf{R}$ is the spectral response function (SRF) that decimates C to c spectral bands. The noise vector $\boldsymbol{\epsilon}$ is distributed as $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Omega})$. Similarly, $\mathbf{N}_g = [\mathbf{n}_{ig}]_{i=1}^c$, where $\mathbf{n}_{ig} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Omega}_{ig})$. The information for constructing $\mathbf{B}$, using the sensors' point spread function (PSF), and $\mathbf{R}$ is either known (i.e., given by the manufacturer) or unknown. In the latter case, $\mathbf{B}$ and $\mathbf{R}$ can be estimated by using the observation data [24]. In this article, we assume that $\mathbf{B}$ and $\mathbf{R}$ are known.

B. Related Work

The common approach for finding a solution for the ill-posed inverse problem (1) and (2) is to solve a regularized least-squares problem, where the regularizer is carefully designed based on the assumed image characteristics. For example, the low-rank assumption is based on the fact that the RS images have a high correlation between the bands. Thus, the target image lives in a low-dimensional subspace. Often, TV is used as a regularizer since it assumes that the image is piecewise smooth and has sharp edges. The nonlocal similarity assumes that the image has repetitive patches across the spatial

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domain. Several model-based methods successfully applied those regularizers for solving inverse problem in RS image fusion [23], [24], [25], [28], [29], [30], [34], [36], to name a few.

Recent developments in DL reveal that a CNN structure is a good image prior (called DIP) [55]. DIP bridges the gap between the model-based and the DL-based methods since it allows one to optimize a CNN-based image reconstruction similar to model-based methods. The main issue of DIP is overfitting. DIP has been improved by building better CNN structures [50], [56], adding explicit constraints [57], [58], and alleviating overfitting by using SURE [59], [60].

Another direction closely related to our method is preprocessing-based methods. The paper [61] proposed an improvement of the CS and MRA pansharpening methods by deblurring the MSI with the Wiener filter and using the deblurred MSI to replace the interpolated MSI in the CS and MRA methods. Backprojection (BP), also known as the linear inverse filter, was used in [60], [62], and [63]. In those methods, a BP that maps an LR observed image to its HR space is incorporated in the fidelity term of the cost function. Although BP is efficient under low noise, it is inefficient under high noise [62], and training a CNN with BP-based DIP loss function [63] has an overfitting problem. To solve the overfitting problem in backprojection DIP (BP-DIP), [60] proposed a loss function based on the generalized SURE [64], [65], which has a similar objective as ours. However, their objective was to solve a superresolution problem for RGB images, while in this work, we focus on RS image fusion for MSI/HSI.

III. Deep SURE-Based Unsupervised CNN for RS Image Fusion

A. Deep Image Prior

DIP [55] demonstrates that the structure of a CNN provides a prior to recover an image generated by the CNN. The idea behind the DIP is that the CNN structure has a higher impedance to unstructured images (e.g., noise) than to structured images. It means that CNN tends to fit an image faster than noise. DIP involves optimizing a CNN using the following loss function

$$\mathcal{L}_{\text{DIP}}(\boldsymbol{\Theta}) = \|\mathbf{y} - \mathbf{H} f_{\boldsymbol{\Theta}}(\mathbf{z})\|_2^2 \tag{3}$$

where $\mathbf{z}$ is the CNN input, which is formed by concatenating the LR image $\mathbf{y}$ and the guided image $\mathbf{g}$, and $f_{\boldsymbol{\Theta}}(\cdot) \in \mathbb{R}^{N\times C}$ is the CNN output. The optimization problem (3) is similar to the one in the model-based methods where the DIP replaces the handcrafted prior. Another idea is the BP-DIP [63], which optimizes a CNN using

$$\mathcal{L}_{\text{BP-DIP}}(\boldsymbol{\Theta}) = \|\mathbf{H}^{\dagger}(\mathbf{y} - \mathbf{H} f_{\boldsymbol{\Theta}}(\mathbf{z}))\|_2^2$$

where

$$\mathbf{H}^{\dagger} = \mathbf{H}^T(\mathbf{H}\mathbf{H}^T)^{-1}. \tag{4}$$

The main issues of DIP and BP-DIP are that its performances depend on the CNN structure and overfitting. To make the DIP and BP-DIP efficient, one should carefully design the CNN structure and use an early stopping schedule to avoid overfitting. However, in a real application, it is hard to choose a proper stopping point because there is no criterion to make a stopping decision.

B. Deep SURE for Image Fusion

To avoid overfitting, we would ideally like to optimize a CNN with a loss function that involves the ground truth $\mathbf{x}$ such as

$$R = \mathbb{E}\|\mathbf{P}(\mathbf{H}\mathbf{x} - \mathbf{H} f_{\boldsymbol{\Theta}}(\mathbf{z}))\|_2^2 \tag{5}$$

where $\mathbb{E}$ is the expectation operator and $\mathbf{P}$ is a linear operator that maps an LR image to its HR space.

Unfortunately, we do not observe $\mathbf{x}$ and therefore cannot compute (5). To overcome this problem, we propose to use a computable unbiased estimator for (5), which is based on SURE. In the following, we derive this estimator. The main goal of this derivation is to eliminate the dependency of $\mathbf{x}$ from (5).

Adding and subtracting $\mathbf{u} = \mathbf{P}\mathbf{y}$ in (4), we have

$$\begin{aligned} R &= \mathbb{E}\|(\mathbf{u} - \mathbf{P}\mathbf{H}\mathbf{x}) - (\mathbf{u} - \mathbf{P}\mathbf{H} f_{\boldsymbol{\Theta}}(\mathbf{z}))\|_2^2 \\ &= \mathbb{E}\|\mathbf{n} - \mathbf{e}\|_2^2 \\ &= \mathbb{E}\|\mathbf{e}\|_2^2 + \mathbb{E}\|\mathbf{n}\|_2^2 - 2\mathbb{E}[\mathbf{e}^T\mathbf{n}] \end{aligned} \tag{6}$$

where $\mathbf{n} = \mathbf{u} - \mathbf{P}\mathbf{H}\mathbf{x} = \mathbf{P}(\mathbf{y} - \mathbf{H}\mathbf{x})$ and $\mathbf{e} = \mathbf{u} - \mathbf{P}\mathbf{H} f_{\boldsymbol{\Theta}}(\mathbf{z})$. Since, from (1), $\mathbf{y}$ is a Gaussian random variable with mean $\mathbf{H}\mathbf{x}$ and covariance $\boldsymbol{\Omega}$, $\mathbf{u}$ is also a Gaussian random variable with mean $\boldsymbol{\mu}_P = \mathbf{P}\mathbf{H}\mathbf{x}$ and covariance $\boldsymbol{\Omega}_P = \mathbf{P}\boldsymbol{\Omega}\mathbf{P}^T$. We also have that $\|\mathbf{e}\|_2^2$ is an unbiased estimate of $\mathbb{E}\|\mathbf{e}\|_2^2$ and

$$\mathbb{E}\|\mathbf{n}\|_2^2 = \mathbb{E}\big[\mathbf{n}^T\mathbf{n}\big] = \text{tr}\big(\mathbb{E}\big[\mathbf{n}\mathbf{n}^T\big]\big) = \text{tr}(\boldsymbol{\Omega}_P)$$

where $\text{tr}(\cdot)$ is the matrix trace. We turn our attention to the term $\mathbb{E}[\mathbf{e}^T\mathbf{n}]$. By using Stein's lemma [53], we get

$$\begin{aligned} \mathbb{E}\big[\mathbf{e}^T\mathbf{n}\big] &= \mathbb{E}\big[\mathbf{e}^T\mathbf{P}(\mathbf{y} - \mathbf{H}\mathbf{x})\big] \\ &= \mathbb{E}\left[\text{tr}\left(\boldsymbol{\Omega}\frac{\partial \mathbf{P}^T\mathbf{e}}{\partial \mathbf{y}}\right)\right] \\ &= \mathbb{E}\left[\text{tr}\left(\boldsymbol{\Omega}\frac{\partial \mathbf{P}^T\mathbf{e}}{\partial \mathbf{u}}\frac{\partial \mathbf{u}}{\partial \mathbf{y}}\right)\right] \\ &= \mathbb{E}\left[\text{tr}\left(\boldsymbol{\Omega}\frac{\partial \mathbf{P}^T\mathbf{e}}{\partial \mathbf{u}}\mathbf{P}\right)\right] \\ &= \mathbb{E}\left[\text{tr}\left(\boldsymbol{\Omega}_P\frac{\partial \mathbf{e}}{\partial \mathbf{u}}\right)\right] \\ &= \mathbb{E}\left[\text{tr}\left(\boldsymbol{\Omega}_P\frac{\partial(\mathbf{u} - \mathbf{P}\mathbf{H} f_{\boldsymbol{\Theta}}(\mathbf{z}))}{\partial \mathbf{u}}\right)\right] \\ &= \text{tr}(\boldsymbol{\Omega}_P) - \mathbb{E}\left[\text{tr}\left(\boldsymbol{\Omega}_P\mathbf{P}\mathbf{H}\frac{\partial f_{\boldsymbol{\Theta}}(\mathbf{z})}{\partial \mathbf{u}}\right)\right]. \end{aligned}$$

Substituting this in (6) and use the fact that, for any random variable $\mathbf{z}$, where $\mathbf{z}$ is an unbiased estimator of $\mathbb{E}[\mathbf{z}]$, we get an unbiased risk estimate for R given by

$$\hat{R} = \|\mathbf{P}(\mathbf{y} - \mathbf{H} f_{\boldsymbol{\Theta}}(\mathbf{z}))\|_2^2 + 2\text{tr}\left(\boldsymbol{\Omega}_P\mathbf{P}\mathbf{H}\frac{\partial f_{\boldsymbol{\Theta}}(\mathbf{z})}{\partial \mathbf{u}}\right) - \text{tr}(\boldsymbol{\Omega}_P).$$

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Dropping $\text{tr}(\mathbf{\Omega}_P)$, which does not depend on the network parameters, we propose an SURE loss function to train (optimize) a CNN as

$$\mathcal{L}_{\text{SURE}}(\mathbf{\Theta}) = \|\mathbf{P}(\mathbf{y} - \mathbf{H}f_{\mathbf{\Theta}}(\mathbf{z}))\|_2^2 + 2\text{tr}\left(\mathbf{\Omega}_P\mathbf{P}\mathbf{H}\frac{\partial f_{\mathbf{\Theta}}(\mathbf{z})}{\partial \mathbf{u}}\right). \quad (7)$$

The trace term in (7) can be viewed as correction for overfitting this can be clearly seen, for example, in Fig. 3. Another insight in the trace term is that it arises from the term $\mathbb{E}[\mathbf{e}^T\mathbf{n}]$. In the supervised setting, $\mathbf{e}$ is a function of training data, and $\mathbf{n}$ is a function of independent validation data. Due to independence, $\mathbb{E}[\mathbf{e}^T\mathbf{n}] = \mathbb{E}[\mathbf{e}^T]\mathbb{E}[\mathbf{n}] = 0$. Therefore, in the supervised setting, the trace term is zero indicating no overfitting.

The trace term in (7) does not have a closed-form expression because the CNN function is highly nonlinear and has too many parameters. It is possible to compute the trace term by using the automatic differentiation of the DL framework [66], e.g., Tensorflow and PyTorch. However, we adopt an approximation approach based on Monte Carlo SURE [67] to approximate the trace term due to its simplicity and fast execution. The Monte Carlo SURE computation is given by

$$\text{tr}\left(\mathbf{\Omega}_P\mathbf{P}\mathbf{H}\frac{\partial f_{\mathbf{\Theta}}(\mathbf{z})}{\partial \mathbf{u}}\right) \approx \mathbf{b}^T\mathbf{\Omega}_P\mathbf{P}\mathbf{H}\frac{f_{\mathbf{\Theta}}(\mathbf{z}+\eta\mathbf{b}) - f_{\mathbf{\Theta}}(\mathbf{z})}{\eta} \quad (8)$$

where $\mathbf{b}$ is drawn from a zero-mean, unit variance Gaussian distribution, and η is a small number. In practice, the noise covariance matrix $\mathbf{\Omega}$ is unknown. However, it can be estimated using the HySure [68] algorithm or the median absolute deviation of the details subband (HH) using the 2-D wavelet decomposition [69].

C. Computation of the BP Operator

In this article, we are interested in using the BP given in (4). We will analyze the performance of $\mathbf{P}$ in Sections IV-B and V-B. With $\mathbf{P} = \mathbf{H}^\dagger$, (7) becomes

$$\mathcal{L}_{\text{SURE}}(\mathbf{\Theta}) = \mathcal{L}_{\text{BP-DIP}}(\mathbf{\Theta}) + 2\text{tr}\left(\mathbf{\Omega}_P\frac{\partial f_{\mathbf{\Theta}}(\mathbf{z})}{\partial \mathbf{u}}\right) \quad (9)$$

since $\mathbf{\Omega}_P\mathbf{H}^\dagger\mathbf{H} = \mathbf{\Omega}_P$, and for this case, the same changes are applied in the calculation of Monte Carlo SURE in (8).

To make the SURE loss function (9) practical, we need fast computation of $\mathbf{P}$ in (4) and the trace term. Computing $\mathbf{P}$ in (4) involves computing a transpose term, $\mathbf{H}^T$, and an inverse term of the blurring and downsampling matrices, $(\mathbf{H}\mathbf{H}^T)^{-1}$. The first term can be interpreted as upsampling by inserting zeroes between samples ($\mathbf{M}^T$) and filtering ($\mathbf{B}^T$) with the blurring kernel $\overline{\mathcal{F}(\mathbf{h})}$, where $\mathbf{h} = [\mathbf{h}^{(i)}]_{i=1}^C$ is the PSF vector of $\mathbf{B}$ and is constructed by stacking the ith band PSF vector, $\mathbf{h}^{(i)}$, on top of each other. Here, $\mathcal{F}(\cdot)$ and $\overline{\mathcal{F}(\cdot)}$ represent the FFT and conjugate FFT, respectively. The second term can be interpreted as the inversion of sequence operators of upsampling ($\mathbf{M}^T$), filtering ($\mathbf{B}\mathbf{B}^T$), and downsampling ($\mathbf{M}$). These sequence operators can be efficiently computed using the polyphase decomposition of the filter $\mathbf{B}\mathbf{B}^T$ [70]. Note that $\mathbf{H}\mathbf{H}^T$ is ill-conditioned (i.e., having many singular values close to zero). It is needed to regularize zero singular values by a small number α, which is manually chosen as a constant of 10^{-3} for all bands. The computation for $\mathbf{P} = \mathbf{H}^\dagger$ applied to an LR image $\mathbf{Y}$ is summarized in Algorithm 1.

Algorithm 1 Fast Computation for the BP $\mathbf{P} = \mathbf{H}^\dagger$ Applied to LR Image $\mathbf{Y}$

Input: LR image $\mathbf{Y} = [\mathbf{y}_i]_{i=1}^C$,
blurring kernel $\mathbf{h} = [\mathbf{h}^{(i)}]_{i=1}^C$,
up/downsampling factor $\mathbf{d} = [d_i]_{i=1}^C$,
regularized value $\boldsymbol{\alpha} = [\alpha_i]_{i=1}^C = 10^{-3}$
Output: Filtered image $\mathbf{X}_F = [\mathbf{x}_{iF}]_{i=1}^C$
/* Repeating for all bands */
for *i=1:C* **do**
  $\tilde{\mathbf{h}}^{(i)} = \mathcal{F}^{-1}\{|\mathcal{F}(\mathbf{h}^{(i)})|^2\}$
  $\mathbf{h}_0^{(i)} =_{(\downarrow d_i)} \tilde{\mathbf{h}}^{(i)}$ // Downsampling by d_i
  $\mathbf{y}_{iF} = \mathcal{F}^{-1}\frac{\mathcal{F}(\mathbf{y}_i)}{\mathcal{F}(\mathbf{h}_0^{(i)})+\alpha_i}$
  $\mathbf{x}_{iUP} = \mathbf{M}_i^T\mathbf{y}_{iF}$ // Upsampling by inserting zeros between samples
  $\mathbf{x}_{iF} = \mathcal{F}^{-1}\{\mathcal{F}(\mathbf{x}_{iUP})\overline{\mathcal{F}(\mathbf{h}^{(i)})}\}$
end

D. Network Structure

To demonstrate our proposed method, we use a fusion CNN with skip connections [55] and attention gates (AGs) [26], [56], [71]. The network architecture is shown in Fig. 1. The network is composed of a guided subnetwork and a fusion subnetwork. The guided image $\mathbf{G}$ is propagated through the guided subnetwork, which is an autoencoder-like network with skip connections to extract multilevel guided feature maps. The guided feature maps merge with the feature maps at the fusion subnetwork using several AGs. Since the autoencoder-like network uses skip connections, the fusion subnetwork requires only the guided feature maps at the decoder to be fused in every AG. Denoting the guided feature maps and the fusion feature maps at layer ith as $\mathbf{z}_g^{(i)}$ and $\mathbf{z}_y^{(i)}$, respectively, the output of each AG is

$$\mathbf{z}_y^{(i+1)} = \text{LeakyReLU}\left(F_g\left(\mathbf{z}_g^{(i)}\right) \odot F_y\left(\mathbf{z}_y^{(i)}\right)\right)$$

where $\odot$ represents an elementwise multiplication. $F_g(\cdot)$ is a function of 128 convolution filters with filter size of 1 ($\text{conv}(1\times 1)\times 128$), followed by a LeakyReLU and a sigmoid layer. $F_y(\cdot)$ is a function of $\text{conv}(1\times 1)\times 128$ followed by a batch normalization (BN) layer. Note that AG is a multiplicative transformation that scales the guided feature maps with different weights. Therefore, the guided feature maps are better aligned with the fusion feature maps yielding a better fusion result [26], [56].

The network is implemented in Pytorch using the Adam optimizer [72] with a learning rate of 0.01.

IV. MS–HS Image Fusion: Experimental Results

Here, we demonstrate the proposed SURE-based unsupervised CNN method for MS–HS fusion. In MS–HS fusion, the guided image is the MSI, and the LR image is the LR HSI. The ideal objective function to be optimized is

$$R_{\text{MH}} = \mathbb{E}\|\mathbf{P}(\mathbf{H}\mathbf{x} - \mathbf{H}f_{\mathbf{\Theta}}(\mathbf{z}))\|_2^2 + \lambda\mathbb{E}\|(\mathbf{F}_{\mathbf{\Theta}}(\mathbf{z}) - \mathbf{X})\mathbf{R}\|_F^2 \quad (10)$$

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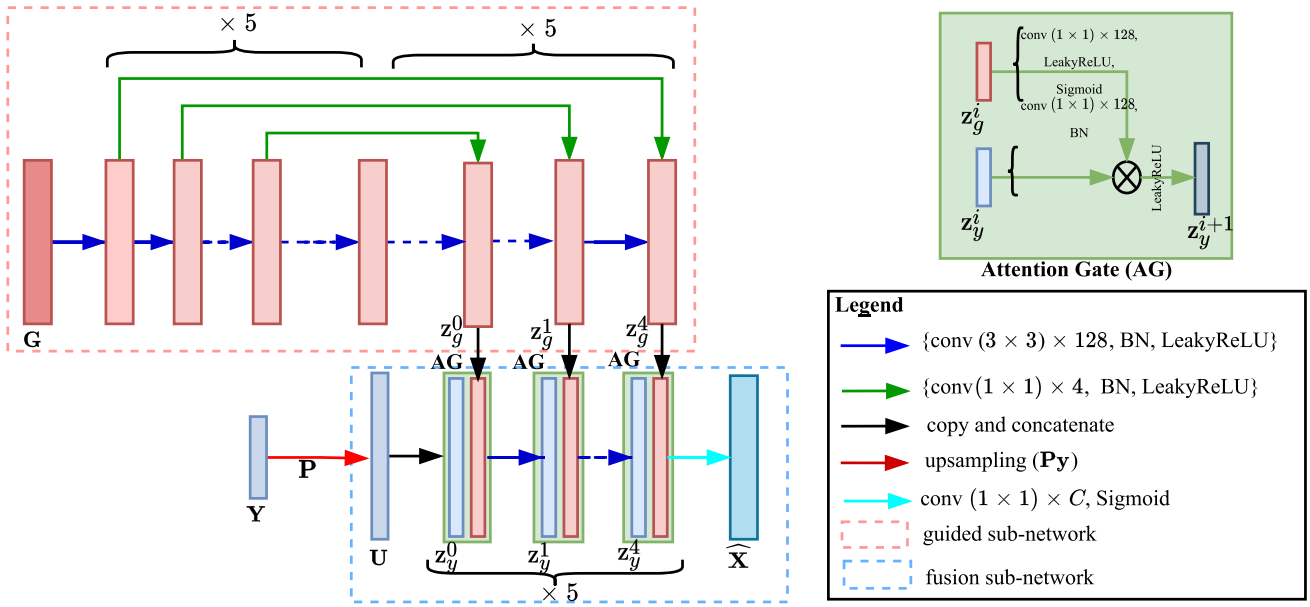

Fig. 1. Fusion CNN architecture. The rectangles represent the images or latent feature maps at the corresponding layers, and ×5 means five times repetition.

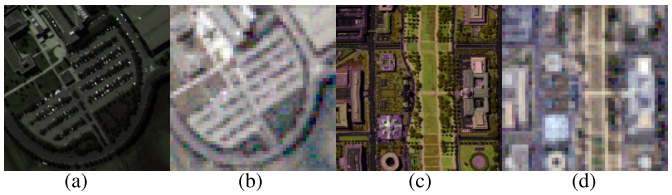

Fig. 2. PU and DC datasets (high noise). The MSIs are shown in false-color images using bands 2, 1, and 3 as the red (R), green (G), and blue (B) channels. The LR images are shown in natural color images using an HSI to RGB color rendering method [73]. (a) PU: MSI. (b) PU: LR HSI. (c) DC: MSI. (d) DC: LR HSI.

where $\mathbf{F}_{\boldsymbol{\Theta}}(\mathbf{z})$ is the CNN output presented in a matrix form and λ is a nonnegative number controlling the tradeoff between two terms. However, $\mathbf{X}$ is unknown, and we exchange (10) to its risk estimate using SURE (9) as follows:

$$\mathcal{L}_{\text{SURE(MH)}}(\boldsymbol{\Theta}) = \hat{R}_{\text{MH}} = \hat{R}_{\text{H}} + \lambda \hat{R}_{\text{M}} \tag{11}$$

where

$$\hat{R}_{\text{H}} = \|\mathbf{P}(\mathbf{y} - \mathbf{H} f_{\boldsymbol{\Theta}}(\mathbf{z}))\|_2^2 + 2\mathbf{b}^T \boldsymbol{\Omega}_P \frac{f_{\boldsymbol{\Theta}}(\mathbf{z} + \eta\mathbf{b}) - f_{\boldsymbol{\Theta}}(\mathbf{z})}{\eta}$$

and

$$\hat{R}_{\text{M}} = \|\mathbf{g} - \text{vec}[\mathbf{F}_{\boldsymbol{\Theta}}(\mathbf{z})\mathbf{R}]\|_2^2 + 2\mathbf{b}^T \boldsymbol{\Omega}_g \text{vec}\left(\frac{\mathbf{F}_{\boldsymbol{\Theta}}(\mathbf{z} + \eta\mathbf{b}) - \mathbf{F}_{\boldsymbol{\Theta}}(\mathbf{z})}{\eta}\mathbf{R}\right).$$

Here, $\mathbf{g}$ is the guided image represented as a vector and $\boldsymbol{\Omega}_g$ is a block-diagonal matrix where each element is $\boldsymbol{\Omega}_{ig}$.

A. Dataset and Evaluation Metrics

We use two semisynthesized datasets, the Pavia University (PU) and the Washington DC Mall (DC). To simulate the data, we follow [24]. First, the data are preprocessed by removing some noisy bands, cropping parts of the original data, and normalizing the intensity values between 0 and 1, resulting in $200 \times 200 \times 93$ and $200 \times 200 \times 191$ data cubes for the PU and DC datasets, respectively. Those data cubes are used as the ground truth (HR HSI). Then, the observed LR HSI is generated by the spatial degradation (1) of the HR HSI by using filtering with a Gaussian low pass filter having a standard deviation $\sigma = 2.0$ and support 15×15 and followed by downsampling with a factor of 4, band by band. The observed MSI is generated by the spectral degradation (2) of HR HSI using the IKONOS SRF [24]. We consider two cases of noise described as follows.

1) *Low Noise:* Isotropic Gaussian noise is added to LR HSI and MSI such that the signal-to-noise ratio (SNR) is SNR = 30 dB and SNR = 40 dB, respectively.
2) *High Noise:* Band-dependent Gaussian noise is added to LR HSI, $\mathbf{n}_i \sim \mathcal{N}(\mathbf{0}, \sigma_i^2\mathbf{I}), i = 1, \ldots, C$, where σ_i is randomly chosen between 0 and 0.1 resulting in SNR = 11.56 dB and SNR = 13.75 dB for the PU and DC datasets, respectively. For the MSI, the additive noise is isotropic Gaussian noise with SNR = 40 dB.

The MSI and LR HSI of the PU and DC datasets are shown in Fig. 2 for the high noise case.

The numerical metrics used to evaluate the fusion performance are the peak-signal-to-noise ratio (PSNR) in decibels, ERGAS, and the spectral angle mapper (SAM) in degree [74]. All the metrics (PSNR, ERGAS, and SAM) are calculated with respect to the original images without extra noise added.

B. Validation of SURE for MS–HS Image Fusion

In this section, we verify that optimizing a CNN with the SURE loss function (11) prevents overfitting. In addition, we evaluate the performance of the linear operator $\mathbf{P}$ using a BP operator ($\mathbf{P} = \mathbf{H}^{\dagger}$) and using an upsampling filtering operator ($\mathbf{P} = \mathbf{H}^T$). Fig. 3 shows the experimental results for the PU dataset where the same CNN as in Fig. 1 is trained

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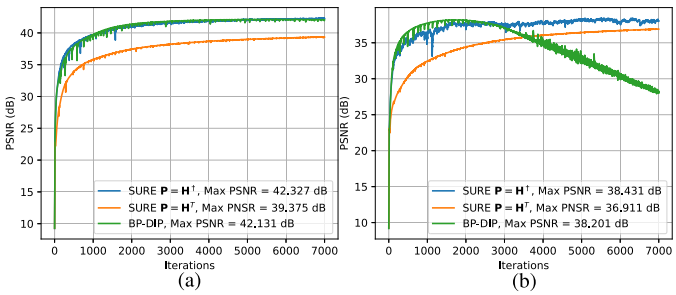

Fig. 3. PSNR curves in training for MS–HS image fusion using the PU dataset. (a) Low noise. (b) High noise.

by using the BP-DIP loss [63], the SURE loss with $\mathbf{P} = \mathbf{H}^T$, and the SURE loss with $\mathbf{P} = \mathbf{H}^\dagger$. The hyperparameter λ is manually chosen as $\lambda = 0.1$ (high noise) and $\lambda = 0.01$ (low noise) for all the loss functions.

BP-DIP significantly overfits in the high noise scenario, as can be seen in Fig. 3(b) where the PSNR curve reaches its peak at 1800 iterations and falls quickly after that. In contrast, the PSNR curves of both SURE loss functions rise and saturate at 4000 iterations, clearly avoiding overfitting. Due to a lack of ground truth in a real application, the PSNR is not computable, and there is no criterion to stop training. In contrast, our proposed SURE-based method is feasible for real applications because it does not depend on an early stopping schedule to avoid overfitting. In a low noise case [Fig. 3(a)], the performance of SURE and BP-DIP losses is similar. This is not surprising since the BP-DIP and SURE losses are different up to a value (the trace term), which depends on the noise variance and is very small in the low noise case. Ideally, BP-DIP and SURE loss functions are exactly the same in the noiseless case.

Fig. 3 also reveals that the performance of the SURE loss with $\mathbf{P} = \mathbf{H}^\dagger$ is better than the SURE loss with $\mathbf{P} = \mathbf{H}^T$ (the average PSNR is about 2.0 dB higher) under the low noise scenario, while they give roughly the same highest PSNR in the high noise case. The reason might be that the linear inverse filter is degraded when the noise level increases causing $\mathbf{H}$ to be more ill-conditioned [62], [63].

C. Performance Evaluation of SURE for MS–HS Fusion

The proposed method is compared to both the model-based and unsupervised DL-based methods, which are described as follows. The first two model-based methods are CNMF [23][1] and SupResPALM [25],[2] which rely on a matrix factorization and unmixing constraints, and the third model-based method, called HySure [24],[3] uses a subspace decomposition and TV regularization. The fourth method is GDD [56],[4] which is an unsupervised CNN using the DIP loss. The performance of HySure and SupRePALM strongly depends on the subspace dimension (k) and the number of endmembers (p). We use a grid search based on PSNR to find the best values for k and p in searching space of $k \in \{3, 5, 7, 10, 15\}$ and $p \in \{3, 4, 5, 6, 7, 10, 15\}$. For low noise, the best parameters of HySure are $k = 10$ and $k = 15$ and the best parameters found for SupResPALM are $p = 15$ and $p = 20$ for the PU and DC datasets, respectively. For high noise, the best parameters of HySure are $k = 3$ and $k = 5$ and the best parameters found for SupResPALM are $p = 4$ and $p = 6$ for the PU and DC datasets, respectively. On the other hand, CNMF is a parameter-free method since it utilizes an automatic algorithm to select tuning parameters. The balance parameter λ in GDD is manually set to $\lambda = 0.1$ for all datasets and noise levels, while balance parameter λ in SURE is empirically chosen as $\lambda = 0.1$ (low noise) and $\lambda = 0.01$ (high noise) for all datasets. Since there is no criterion to stop training GDD, we follow a strategy in [55] where the fused image is the exponential average of the CNN output with a weight of 0.99, and the number of iterations is chosen manually as 3000 iterations for both datasets. The same strategy used to obtain the fused image in GDD is also applied to our proposed method for a fair comparison. We want to emphasize that we allow the competitive model-based methods to use the ground truth to find the best parameters, while our proposed method does not rely on the ground truth. Thus, our proposed method is more practical than the competitive model-based methods.

[1]http://naotoyokoya.com/Download.html
[2]https://github.com/lanha/SupResPALM
[3]https://github.com/lanha/SupResPALMhttps://github.com/alfaiate/HySure
[4]https://github.com/tuezato/guided-deep-decoder

TABLE I

MS–HS IMAGE FUSION RESULTS FOR THE PU DATASET IN TERMS OF PSNR (DB), ERGAS, AND SAM (°). THE DOWN ARROW MEANS THAT SMALL VALUES ARE BETTER, WHILE THE UP ARROW MEANS THAT LARGE VALUES ARE BETTER. THE BEST RESULTS ARE SHOWN IN BOLD

Low noise SNR (MSI)=40 dB, SNR (HSI)=30 dB					
	CNMF	HySure	SupRes PALM	GDD	SURE
PSNR(dB)↑	37.17	40.62	30.62	40.14	**41.57**
ERGAS↓	2.193	1.460	4.612	1.511	**1.337**
SAM(°)↓	2.783	2.317	4.028	2.457	**1.959**
High noise SNR (MSI)=40 dB, SNR (HSI)=11.56 dB					
	CNMF	HySure	SupRes PALM	GDD	SURE
PSNR(dB)↑	32.38	34.85	29.51	36.27	**38.47**
ERGAS↓	3.854	3.209	5.267	2.35	**1.867**
SAM(°)↓	5.605	4.041	5.349	4.195	**3.185**

Tables I and II show the results for the PU and DC datasets, respectively. Our proposed method gives better results than the competitive methods in all metrics for both datasets. In general, the unsupervised DL-based methods perform better than the model-based methods. However, it is noticeable that HySure works comparably as well as SURE and better than the GDD method in terms of PSNR for the PU dataset under the low noise scenario. Figs. 4 and 5 show the fused images obtained by all methods in the high noise case for the PU and DC datasets, respectively. For the PU datasets,

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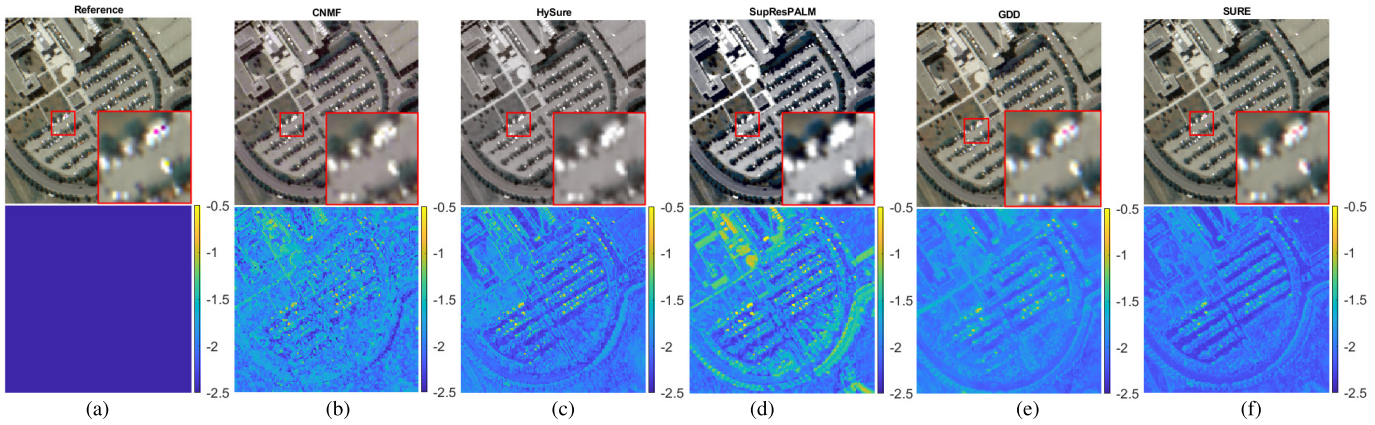

Fig. 4. Fusion results for the PU dataset (high noise). The first row is the images shown in natural color using an HSI to RGB color rendering method [73]. The images shown in big red rectangles are the four times zooming in of the images shown in small red rectangles. The second row is the RMSE-based residual images (shown in the logarithm scale) with respect to the reference. (a) Reference. (b) CNFM. (c) HySure. (d) SupResPALM. (e) GDD. (f) SURE.

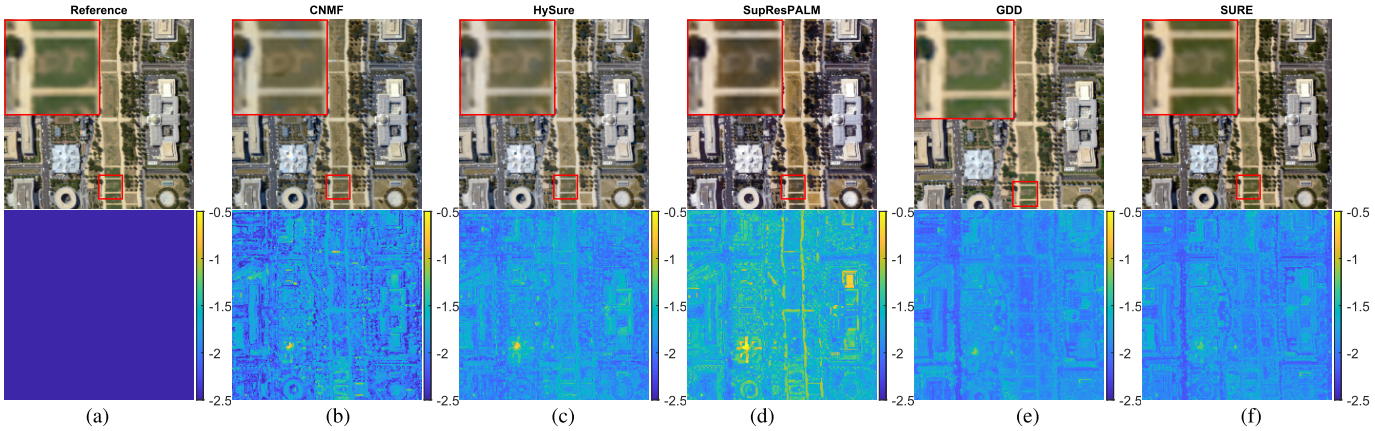

Fig. 5. Fusion results for the DC dataset (high noise). The first row is the images shown in natural color using an HSI to RGB color rendering method [73]. The images shown in big red rectangles are the four times zooming in of the images shown in small red rectangles. The second row is the RMSE-based residual images (shown in the logarithm scale) with respect to the reference. (a) Reference. (b) CNFM. (c) HySure. (d) SupResPALM. (e) GDD. (f) SURE.

although the numerical result of the proposed method is better than the GDD method, the fused images are not noticeably different and they are closer to the reference than the other methods. HySure yields a slightly blurry image, while the SupResPALM images seem spectrally distorted. For the DC dataset, the difference in the results of CNMF, GDD, and our proposed method is not visually pronounced, and the fused images by those methods look closer to the reference than the other methods. Further inspection of the root-mean-square error (RMSE)-based residual images (i.e., the RMSE computed along the third dimension of an HSI image) of the estimated HR images with respect to the reference images tells us that the proposed method yields less residual structure than the competitive methods. In contrast, SupResPALM yields the highest residual structure among all methods.

V. MS–MS Image Fusion: Experimental Results

Here, we demonstrate our proposed SURE method for MS–MS image fusion. Representative examples of MS–MS image fusion are S2 sharpening and pansharpening.

The S2 satellite imaging system provides 13 MS bands, comprising four bands at 10-m (spatial) resolution, six at 20-m resolution, and three at 60-m resolution. We ignore one 60-m band (band 10) since it does not contain the Earth's surface information. The S2 sharpening problem is to estimate the 20- and 60-m bands at 10-m resolution by fusing them with the 10-m bands. In S2 sharpening, there is no explicit relationship between the guided image (the 10-m bands) and the estimated LR image (the 20- and 60-m bands). The guided image is assumed as a part of the fused image so that the linear mapping operator $\mathbf{P}$ applied to the guided image part is identity. Although we are interested in estimating the LR images at their maximum resolution, adding the guided image to the fused image improves the fusing results since it exploits the correlation between all the S2 bands [51]. To apply our proposed method for the S2 sharpening, we used the CNN in Fig. 1 optimized with the SURE loss function (7).

A. Datasets and Evaluation Metrics

The proposed method is evaluated using simulated and real S2 datasets. The simulated dataset is an HSI collected by the Airborne Prism Experiment (APEX). We follow the procedure in [25] to create the APEX MSI having 12 bands of 198 ×

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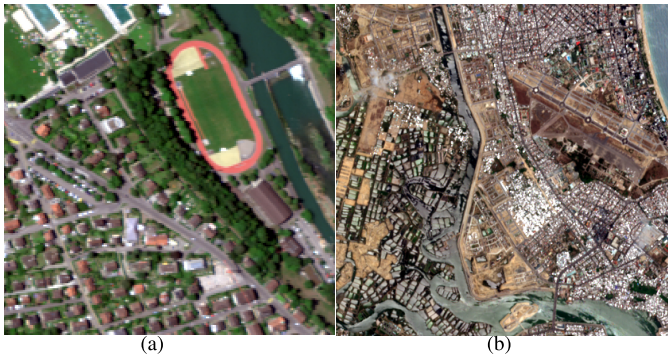
TABLE II

MS–HS IMAGE FUSION RESULTS FOR THE DC DATASET IN TERMS OF PSNR (DB), ERGAS, AND SAM (°). THE DOWN ARROW MEANS THAT SMALL VALUES ARE BETTER, WHILE THE UP ARROW MEANS THAT LARGE VALUES ARE BETTER. THE BEST RESULTS ARE SHOWN IN BOLD

Low noise SNR (MSI)=40 dB, SNR (HSI)=30 dB					
	CNMF	HySure	SupRes PALM	GDD	SURE
PSNR(dB)↑	36.41	35.35	30.02	37.43	**38.81**
ERGAS↓	1.847	1.828	4.287	1.663	**1.480**
SAM(°)↓	3.861	4.333	5.926	3.700	**3.265**
High noise SNR (MSI)=40 dB, SNR (HSI)=12.97 dB					
	CNMF	HySure	SupRes PALM	GDD	SURE
PSNR(dB)↑	33.34	34.58	28.71	36.08	**36.21**
ERGAS↓	3.135	2.527	4.968	2.192	**1.994**
SAM(°)↓	5.430	5.237	6.928	4.547	**4.446**

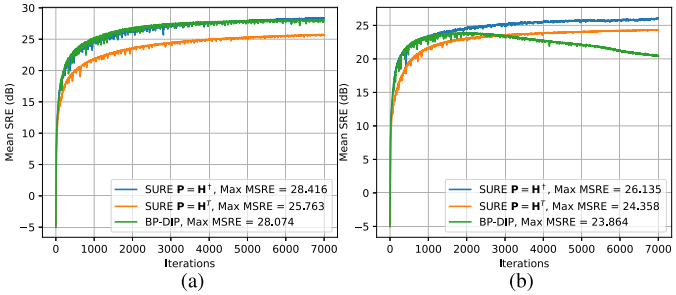


(a) (b)

Fig. 6. S2 simulated and real datasets. The images are shown in natural color using bands 2, 3, and 4 as the R, G, and B channels. (a) APEX (SNR = 40 dB). (b) Sub-urban.

198 pixels per band and a resolution of 2 m. To simulate the 20- and 60-m bands, we apply a spatial filter to the reference with the PSFs taken from the S2 sensor's specifications [75] followed by a downsampling operator with a factor of 2 and 6, respectively. Isotropic Gaussian noise is added to the observed data such that the LR and noisy images have SNR = 30 dB and SNR = 40 dB for all bands. The real S2 dataset is a level 1C product and can be downloaded from the ESA/Copernicus portal.[5] This real S2 dataset (called Sub-urban) shows a part of a suburban area in Khanh Hoa province, Vietnam, and was acquired in April 2019. The spatial size of the 10-m bands of the Sub-urban dataset is 420 × 420 pixels. Fig. 6 shows the 10-m bands of the APEX (SNR = 30 dB) and Sub-urban datasets in natural color using bands 2, 3, and 4 as the R, G, and B channels.

[5]https://scihub.copernicus.eu/dhus/#/home



(a) (b)

Fig. 7. MSRE curves in training for the APEX dataset using different loss functions. (a) SNR = 40 dB. (b) SNR = 30 dB.

For the APEX dataset, the following evaluation metrics are used, the mean signal-to-reconstructed error (MSRE) in decibels [30], the SAM in degree, and the mean structure similarity index (MSSIM) [30]. All the metrics (MSRE, SAM, and MSSIM) are calculated with respect to the original images without extra noise added. We assess the results visually for the Sub-urban dataset due to the lack of reference images.

B. Validation of SURE for S2 Sharpening

Here, we verify the advantage of the SURE loss function in terms of overfitting avoidance and the SURE loss's performance with different linear operators $\mathbf{P}$. To demonstrate that, we use the same CNN as in Fig. 1 optimized with the BP-DIP [63], the SURE loss function with $\mathbf{P} = \mathbf{H}^{\dagger}$ and with $\mathbf{P} = \mathbf{H}^{T}$.

Fig. 7 shows the MSRE curves in training using the three loss functions mentioned above and the APEX dataset with SNR = 40 dB and SNR = 30 dB. The same trend as in the MS–HS image fusion presented in Section IV-B is observed. There, the BP-DIP loss function has an overfitting problem, specifically in the higher noise case [Fig. 7(b)], while the SURE loss functions do not have it. The performance of the SURE loss with $\mathbf{P} = \mathbf{H}^{\dagger}$ is significantly better than one with $\mathbf{P} = \mathbf{H}^{T}$ when the noise level decreases.

C. Performance Evaluation of SURE for S2 Sharpening

This section presents the performance evaluation of SURE for S2 sharpening using the simulated APEX dataset and the real Sub-urban dataset. In addition, we compare the proposed SURE-based method against the recently published S2 sharpening methods. The competitive methods are three model-based methods (SupReME [28],[6] S2Sharp [30],[7] and SSSS [31][8]) and one unsupervised DL-based method that is our proposed method trained with the BP-DIP loss function [63]. SupReME and S2Sharp exploit the low-rank property of the S2 data, where the fused images are the solutions to the penalized least-squares problems in a reduced-rank domain with a roughness regularizer. The difference between SupReME and S2Sharp is that while SupReME assumes that the parameters (e.g., the dimension of subspace and the

[6]https://github.com/lanha/SupReME
[7]https://github.com/mou12/S2Sharp
[8]https://sites.google.com/view/chiahsianglin/software

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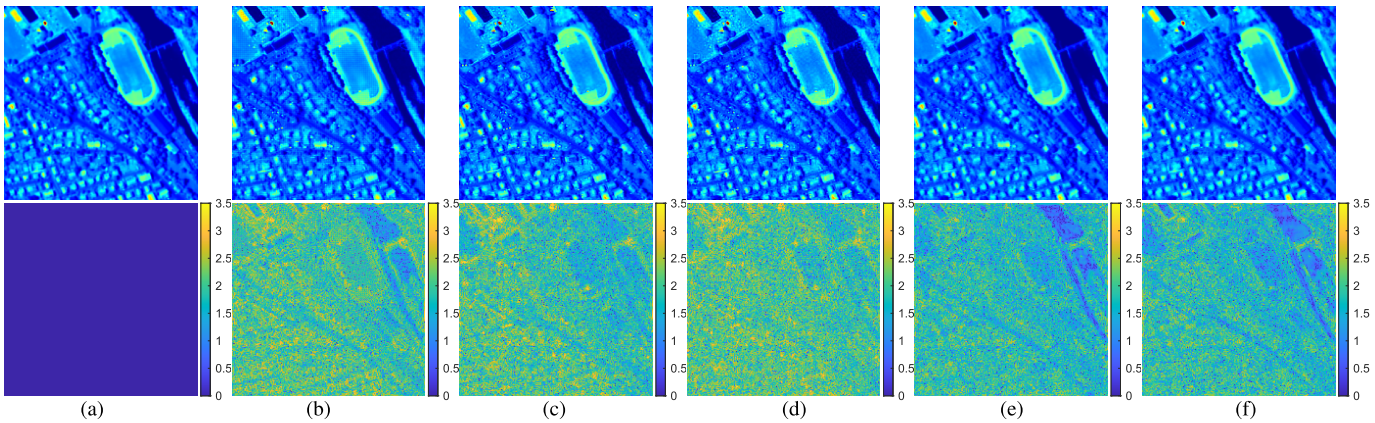

Fig. 8. Image fusion results for the APEX dataset (SNR = 40 dB). (Top row) Images for band 11 (20 m). (Bottom row) Respective residual images shown in the logarithm scale. (a) Reference. (b) SupReME. (c) S2Sharp. (d) SSSS. (e) BP-DIP. (f) SURE.

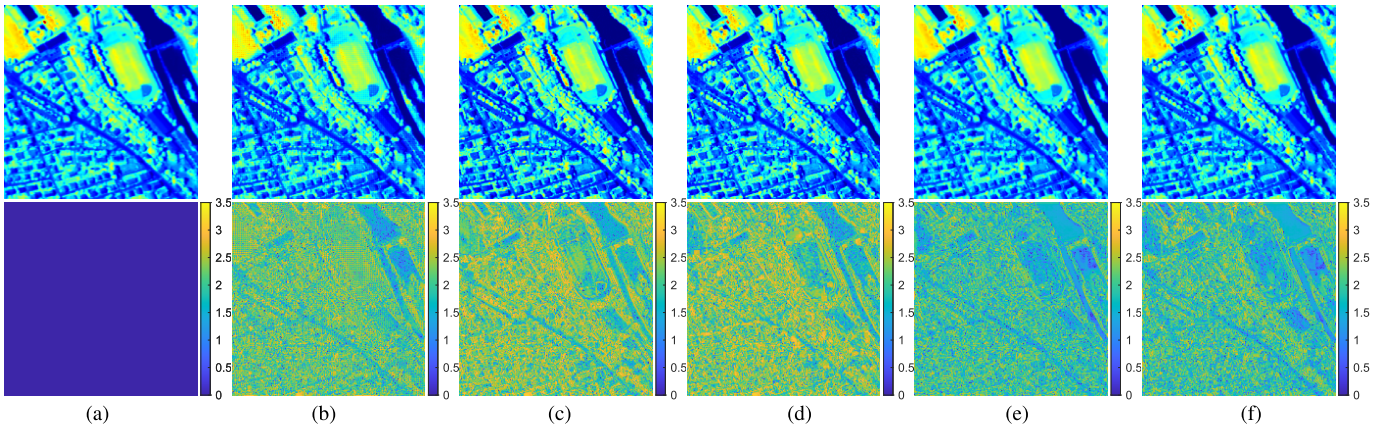

Fig. 9. Image fusion results for the APEX dataset (SNR = 40 dB). (Top row) Images for band 9 (60 m). (Bottom row) Respective residual images shown in the logarithm scale. (a) Reference. (b) SupReME. (c) S2Sharp. (d) SSSS. (e) BP-DIP. (f) SURE.

Fig. 10. Image fusion results of the 20-m bands for the Sub-urban dataset. The images are shown in false-color images using bands 12, 8a, and 5 as the R, G, and B channels. The images shown in big red rectangles are the four times zooming in of the images shown in small red rectangles. (a) Observed 20-m bands. (b) SupReME. (c) S2Sharp. (d) SSSS. (e) BP-DIP. (f) SURE.

regularization weights) are fixed, S2Sharp automatically estimates those parameters using a Bayesian optimization method. SSSS also relies on the low-rank property of the S2 data, but it solves the reduced-rank least-squares problem with the self-similarity regularizer. We experimentally observed that the main parameters affecting the results of SupReME and SSSS are the dimension of subspace k and the regularization weight λ. Using a grid search for the APEX dataset, the best parameters (based on highest MSRE) of SupReME and SSSS for both SNR = 30 dB and SNR = 40 dB were found to be $k = 8$ and $\lambda = 0.01032$ and $k = 8$ and $\lambda = 0.01$, respectively. Since there is no criterion to stop training the CNN using BP-DIP loss, we manually stop optimizing the CNN at 3000 iterations, and the result is the running average of the CNN output with an exponential weight is 0.99, as a recommendation in [55]. For a fair comparison, the same schedule to obtain a result in BP-DIP is also applied in our proposed method.

The quantitative results are shown in Table III for the APEX dataset. In the low noise scenario of SNR = 40 dB, BP-DIP and SURE yield similar results as was discussed in Section V-B. However, in a higher noise scenario of

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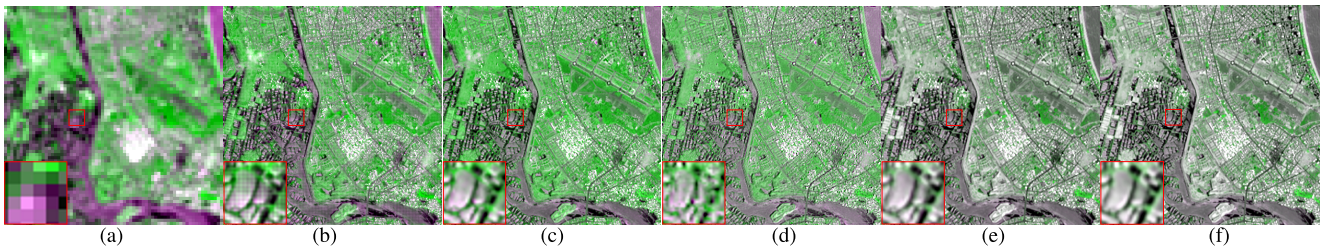

Fig. 11. Image fusion results of the 60-m bands for the Sub-urban dataset. The images are shown in false-color images using bands 1, 9, and 1 as the R, G, and B channels. The images shown in big red rectangles are the four times zooming in of the images shown in small red rectangles. (a) Observed 60-m bands. (b) SupReME. (c) S2Sharp. (d) SSSS. (e) BP-DIP. (f) SURE.

TABLE III

S2 SHARPENING RESULTS (60- AND 20-M BANDS) FOR THE APEX DATASET IN TERMS OF MSRE (DB), SAM (°), AND MSSIM. THE DOWN ARROW MEANS THAT SMALL VALUES ARE BETTER, WHILE THE UP ARROW MEANS THAT LARGE VALUES ARE BETTER. THE BEST RESULTS ARE SHOWN IN BOLD

	SNR=40 dB				
	SupReME	S2Sharp	SSSS	BP-DIP	SURE
MSRE(dB)↑	22.86	26.89	25.43	**27.58**	27.53
SAM↓	3.855	4.250	4.914	**2.421**	2.533
MSSIM↑	0.921	0.937	0.911	**0.955**	**0.955**
Time (s)↓	**3.17**	11.60	182.84	277.01	392.54
	SNR=30 dB				
MSRE (dB)↑	22.34	23.13	20.60	23.72	**25.85**
SAM↓	4.499	5.637	9.033	3.908	**2.893**
MSSIM↑	0.905	0.896	0.836	0.906	**0.940**
Time (s)↓	**3.28**	10.01	180.15	273.06	390.16

SNR = 30 dB, SURE is considerably better than BP-DIP since BP-DIP overfits at 3000 iterations. Both SURE and BP-DIP outperform the model-based methods in all metrics and noise cases. For example, the proposed method is roughly 1.5 and 2.7 dB in terms of MSRE higher than the best model-based method, S2Sharp, for SNR = 40 dB and SNR = 30 dB, respectively. The images sharpened by BP-DIP and SURE have remarkably less spectral distortion than the model-based methods, with 1.2°–6.2° lower in terms of SAM for all noise cases. Figs. 8 and 9 show the sharpened bands 11 (20-m band) and 9 (60-m band), and their residual images (in logarithm scale) of the APEX dataset obtained by all methods and the reference images. Zooming in the reconstructed images of band 11 reveals that SupReME generates some grid-like artifacts, and the images generated by SSSS and S2Sharp are slightly noisy (see the top-left corner). The images of BP-DIP and SURE methods are similar and very close to the reference images. Band 9 has high contrast, and it is very hard to differentiate between the sharpened images visually. However, the residual images of bands 11 and 9 indicate that SURE and BP-DIP yield less residual structure than the competitive methods.

Table III gives the running time of all methods in seconds. The proposed SURE-based method was implemented in Pytorch 1.8 GPU and ran on a Linux computer with eight cores Intel CPU 3.2 GHz, 64 GB of RAM, and 12-GB memory of GPU (Nvidia Titan X). The competitive methods were implemented using MATLAB R2019b and ran on the same computer. The fastest to slowest methods are ranked as SupReME, S2Sharp, SSSS, BP-DIP, and SURE.

The experimental results for the real S2 dataset (Sub-urban) are shown in Figs. 10 and 11. Here, the parameters for all methods are set as the same as in the simulated case. For the 20-m bands shown in Fig. 10, SupReME produces some grid-like artifacts, the remaining methods yield well-looking images, and the difference between those images is not visually pronounced. For the 60-m bands shown in Fig. 11, the SupReME image has some artifacts, and the SSSS image is noisy and blurry. S2Sharp, BP-DIP, and our proposed methods give good sharpened images, which are sharper than SupReME and SSSS images. The results of BP-DIP and SURE are very similar, and this is because the noise might be small for this dataset. Note that BP-DIP and SURE seem to generate spectral distortion in the 60-m bands sharpened images since the color of those images deviates from the observed bands.

D. Pansharpening

In pansharpening, the guided image is a PAN, and the LR image is an MSI. Since the PAN is a linear combination of the bands of an HR MSI, pansharpening can be considered a special case of the MS–HS image fusion where the PAN and LR MSI play the roles of the LR HSI and MSI, respectively. The proposed SURE method is applied to pansharpening by optimizing a CNN with the SURE loss function (11), where the LR HSI and MSI are replaced by the MSI and the PAN, respectively.

We use the simulated Pleiades dataset [10] to evaluate the performance of our proposed method. The Pleiades dataset is collected by an aerial platform, showing a scene of Toulouse (France) in 2006. It has four MS bands (HR MSI) with 60-cm resolution and lacks a PAN image. The LR MSI is simulated by bandwise filtering the HR MSI using a Gaussian low-pass filter with a standard deviation of 2.0 and kernel support of 15 pixels and is downsampled with a factor of 4. The simulated PAN is the average of all bands of the HR MSI. The proposed method is compared against the representative CS, MRA,

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Fig. 12. Pansharpening results for the Pleiades dataset. The reference, LR-MSI, and the pansharpened images of 256×256 pixels are shown in natural color using bands 1, 2, and 3 as the R, G, and B channels. The PAN image is shown in grayscale. The number presented in brackets is the SAM in degrees and ERGAS where the best results are in bold. (a) Reference. (b) LR MSI. (c) PAN. (d) BDSD (5.08°, 2.93). (e) MTF-GLP (5.45°, 4.58). (f) PNN (5.59°, 4.99). (g) SURE (2.76°, 1.74).

and supervised DL-based methods, i.e., BDSD [14], MTF-GLP [19],[9] and PNN [43].[10] For the BDSD and MTF-GLP methods, we modify the modulation transfer function (MTF)-based filter by using the Gaussian filter described above (i.e., a standard deviation of 2 and a kernel support of 15 pixels). For the PNN method, we use the pretrained network provided by the authors.

Experimental results are shown in Fig. 12. The SURE result is significantly better than the competitive methods in quantitative metrics and the pansharpened images' quality. SURE yields 2.76° and 1.74 in terms of SAM and ERGAS that are roughly 2.3° and 1.2 lower than the second best method, BDSD. The competitive methods suffer some degrees of spectral and spatial distortion where the resulting images look blurrier, and the color deviates far from the reference image. It is not surprising that PNN performs poorly, although it is a supervised DL-based method since the pretrained CNN in PNN is trained using the bicubic degradation model that does not match the degradation model assumed in this article.

VI. Conclusion

This article proposed a general framework for optical RS image fusion, taking advantage of the DIP provided by a CNN structure and SURE. The proposed method is a hybrid model and DL-based method that overcomes the difficulty in the handcrafted designed image prior to the model-based methods and the overfitting problem occurring in the unsupervised DIP-based methods. We demonstrated the proposed method with two RS fusion examples: MS–HS and MS–MS fusion. Our proposed method uses a linear inverse filter embedded with SURE, leading to significant improvement results over competitive methods. However, there are still open questions that need to be addressed in the future works, for example, different kinds of preprocessing operators $\mathbf{P}$ (e.g., a nonlinear operator such as a neural network mapping function) and the more efficient way to compute the trace of the derivative of the network with respect to its input. Also, applying the proposed method to the related problems, such as deblurring and compressed sensing will be addressed in our future works.

[9]Codes for BDSD and MTF-GLP are available at https://openremotesensing.net/knowledgebase/a-critical-comparison-among-pansharpening-algorithms/

[10]https://github.com/sergiovitale/pansharpening-cnn-MATLAB-version

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